# Health Technology Ecosystem

## Snowflake Response to Request for Information (CMS-0042-NC)

To

## Centers for Medicare & Medicaid Services

## Department of Health and Human Services

June 16, 2025

snowflake®

June 16, 2025

Department of Health and Human Services
Centers for Medicare & Medicaid Services

**Subject:** Snowflake Response to CMS Request for Information for Health Technology Ecosystem

Dear CMS Team,

Snowflake is pleased to submit our response to your Request for Information regarding building a more connected, data-driven digital healthcare ecosystem. We commend CMS's proactive efforts to advance a more connected and efficient healthcare ecosystem, focused on improving patient care, reducing administrative burden, and driving system-wide efficiency.

As of September 2024, nearly 70 million individuals are enrolled in Medicare. Notably, 80% of all Medicare beneficiary data resides in Snowflake today, housed within our extensive network of payer customers. With our secure, multi-cloud data platform and proven data sharing capabilities, we believe Snowflake is uniquely positioned to serve as the data backbone for real-time, point-of-care insights across the digital health landscape.

CMS's initiative is organized around five key pillars. Snowflake's formal response will focus on Pillar 4: Data at the Point of Care and Pillar 5: Enhancing Trusted Data Exchange. The majority of the payer landscape already relies on Snowflake as the trusted platform to manage their most sensitive data, insights, and AI workloads. This RFI represents a natural extension of that trust and a unique opportunity to help build the future of healthcare data interoperability.

To further demonstrate industry alignment with this vision, Snowflake will publish an open letter endorsed by leading payers, expressing strong support for Snowflake's role as a foundational enabler of secure data sharing in support of CMS's digital health interoperability initiative.

Snowflake has a long-standing partnership with CMS, supporting your efforts to modernize core data infrastructure on our platform. With your support, we believe we can further enable CMS's vision of timely, actionable data at the point of care.

Snowflake submits this response for informational purposes only and with the understanding that it will not create any legally binding obligations on either party, and that binding obligations will arise only upon execution of definitive agreements that are mutually acceptable to both parties. However, the information in this response does constitute Snowflake's intellectual property and confidential information and CMS may only use it to evaluate the Snowflake offerings and may not share it.

Thank you for your review and consideration of our response and Snowflake's platform to help meet your needs. If you have any further questions, please feel free to contact your Snowflake account representative any time.

Sincerely,

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# Snowflake Response to CMS RFI

## CMS - Current Data landscape

Centers for Medicare & Medicaid Services is one of Snowflake's largest public sector customers with dozens of individual workloads leveraging the Snowflake AI Data Cloud. Both Medicaid and Medicare claims for Parts A, B, C, D, and Durable Medical Equipment (DME) are hosted in Snowflake in enterprise data platforms. Additionally, Centers for Medicare & Medicaid Services has fully embraced data sharing between systems within the agency to reduce friction & make data available real time and seamlessly where needed.

## The Snowflake AI Data Cloud Platform

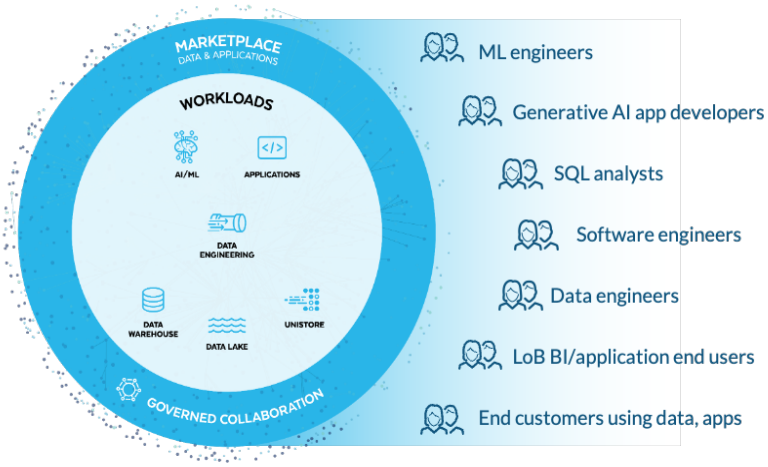
The Snowflake Data Cloud Platform is a single, fully managed platform that powers the AI Data Cloud, securely connecting businesses globally across any type or scale of data to productize AI, applications and more in the enterprise. Our robust, fully featured platform processes enterprise-wide data, including data warehousing, data lake, data engineering, data science, and data sharing capabilities. Founded in August 2012, Snowflake pioneered a cloud-native, SQL-compliant, relational analytical database platform. This fully managed software as a service (SaaS) was unprecedented. Snowflake disrupted legacy data architectures and set a new standard for building data platforms and databases.

The Snowflake AI Data Cloud enables companies to unlock the full value of their enterprise data and use it to deliver transformative AI powered insights, applications and business outcomes. Snowflake makes enterprise AI easy, efficient and trusted with over 11,000 companies around the globe, including hundreds of the world's largest, using Snowflake's AI Data Cloud to share data, build applications, and power their business with AI. The era of enterprise AI is here.

***Data, AI and apps for your enterprise and your end customers***

### Why the AI Data Cloud:

- Unified data and AI strategy
- Fully-managed infrastructure – simplicity, scale, governance
- Collaborate, distribute, and monetize with over 11,000 customers & 2.4K+ Marketplace Listings through the AI Data Cloud




**Snowflake Cortex** is a comprehensive suite of AI features within the Snowflake AI Cloud Platform, designed to empower users to leverage large language models (LLMs) and machine learning directly on their data, without moving it out of Snowflake's secure environment. It offers functionalities like Cortex AISQL for analyzing multi-modal data with SQL, Cortex Analyst for conversational insights, and Cortex Search for efficient retrieval.

Snowflake Cortex integrates with and provides access to a variety of industry-leading LLMs, giving users flexibility and choice. Our key LLM partners include Anthropic (with models like Claude and Claude Sonnet), Meta (including the Llama series), and Mistral AI (with models such as Mistral Large and Mixtral 8x7B). Additionally, Snowflake has developed its own enterprise-focused LLM, Snowflake Arctic, and supports other models like Google's Gemma and DeepSeek. This diverse selection allows organizations to choose the most suitable model for their specific AI applications and use cases.

## Data Sharing

Data sharing has evolved dramatically over the past decade, shifting from cumbersome file-based transfers with compromised governance to streamlined, secure, real-time solutions.

A key architectural component of the Snowflake AI Data Cloud Platform is its ability to securely share data and application assets between Snowflake accounts without the need to move or copy the underlying data. This feature allows seamless, secure, and near real-time collaboration between different teams, departments, or external partners, regardless of the choice of cloud provider or region.

At its core, Snowflake data sharing enables:

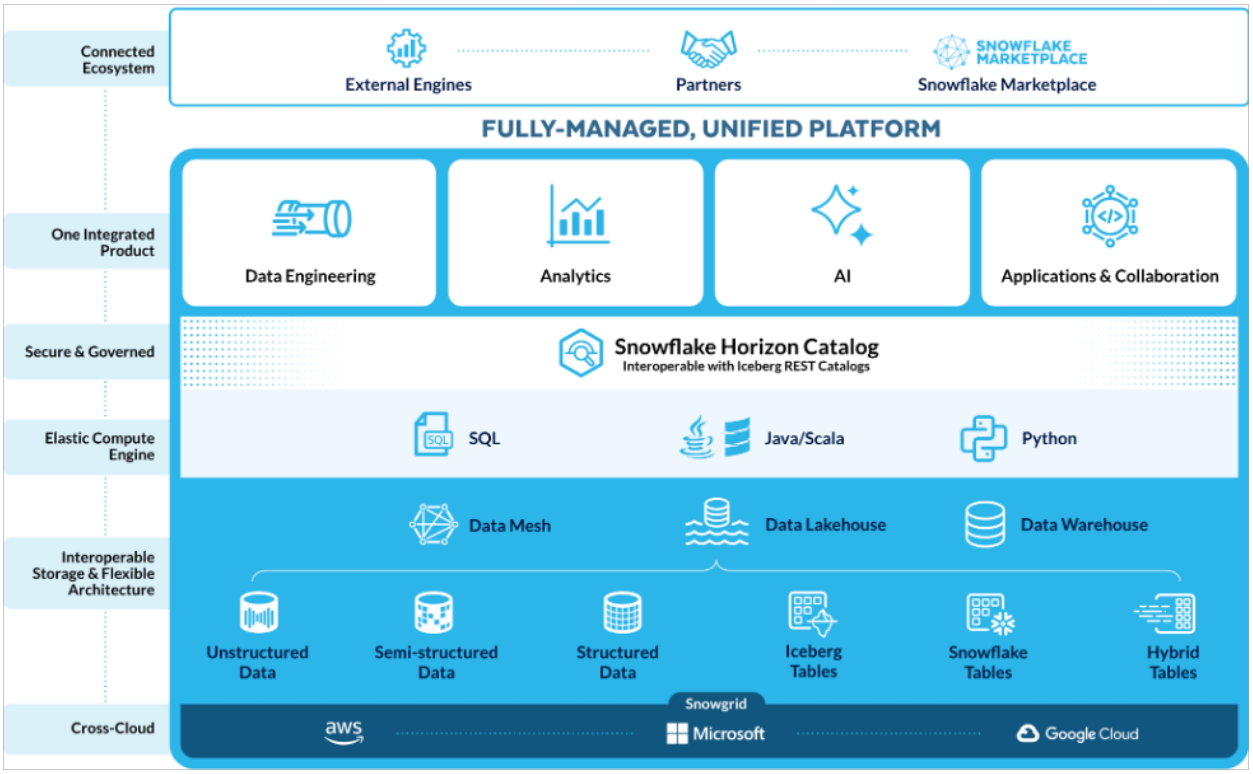
- **Secure Access:** Data remains in the provider's account, and access is granted to consumer accounts via "shares", which are essentially containers pointing to the data. This eliminates the need for data duplication and transfers, enhancing security and maintaining data integrity.
- **Real-time Collaboration:** Consumers have live, read-only access to the shared data, allowing them to query it directly within their own Snowflake account and receive immediate updates as the data changes.
- **Cost Efficiency:** Both providers and consumers benefit from cost savings. Providers avoid the costs of duplicating and transferring data, while consumers only pay for the compute resources (virtual warehouses) they use to query the shared data.
- **Flexibility:** Data can be shared internally (within the same account) or externally (across different Snowflake accounts, even across different cloud providers and regions). Snowflake also offers reader accounts for sharing data with external users who are not Snowflake customers.

Snowflake enables data sharing through the use of Shares, which are account-level objects created by data providers and imported by data consumers. When a data consumer imports a share, data is not moved or copied. The consumer is being granted direct read-only access to the objects that the provider has authorized for them. All the data remains securely stored in the provider’s account, while different users or organizations access it in real-time. This method eliminates data duplication, reduces storage costs, and ensures that all parties work with the most current version of the data.

Snowflake offers several options for sharing data, including Direct Shares, Listings, and Data Exchanges. Direct Shares enable providers to directly share specific database objects with another account in their region. A Listing allows providers to enhance the contents of the share with additional metadata, such as a data dictionary, sample queries, and provider contact information. The listing can be posted to any other Snowflake account, regardless of cloud or region, as a data product. For sharing with wider audiences, Snowflake offers both a public Marketplace and private Data Exchanges. Listings in the Public Marketplace can be seen and imported by any Snowflake customer across the globe, while Private Exchanges are restricted to invited participants only.

Snowflake Data Sharing reduces the friction between data providers and consumers, increases the security and governance of the data assets, and can shorten the time to insight for data-driven organizations.

### One Platform to Unlock AI & Apps Innovation




## Current State - Challenges and Opportunities

Historically, HL7/FHIR servers were sufficient to share the volume of data in the healthcare ecosystem and the fee for service/transactional model. Today, with the implementation of value-based care models requiring a longitudinal patient record and the explosion of data volumes and data types; from genomics to device to audio, video, etc., FHIR servers will not be sufficient. Furthermore, the implementation of FHIR servers today creates a copy of data and therefore data silos. We need to bring data together without making copies, modern cloud technologies provide such a solution.

### A Future State Healthcare Network

We envision a future in which ADT messages still flow to transition patient care from inpatient to outpatient, medication histories are reconciled, and TEFCA facilitates the flow of information (copies) throughout the ecosystem to support transactions. We believe that in order for value care-based models to ultimately succeed, alongside that transactional flow is a “data sharing” flow which will be covered in detail below.

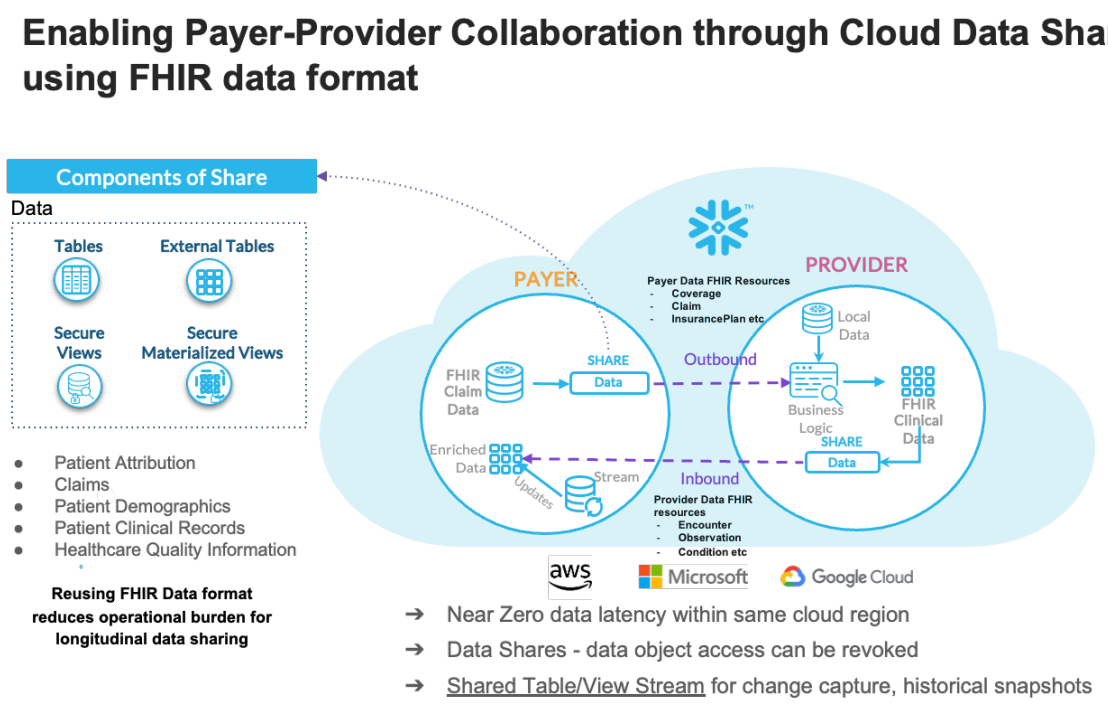
## Leveraging Cloud Data Sharing to Advance CMS's Vision for Interoperable Healthcare Data Particularly for Longitudinal Patient Data Sharing

Modern cloud technologies come with capabilities such as Data Sharing that enables organizations to securely share live data across multiple cloud platforms without the need for traditional data movement or copying. Through a provider-consumer model, organizations can share specific tables, views, or entire databases while maintaining real-time access to the data. What makes this particularly efficient is that consumers can access shared data instantly without requiring complex ETL processes, and they only incur compute costs rather than storage costs, as the shared data remains read-only for consumers.

When it comes to handling FHIR data, the ability to handle semi-structured data formats like JSON seamlessly is very important. FHIR resources, which are fundamentally JSON-based, can be directly loaded into tables and queried VARIANT data type-like capabilities. This capability, combined with built-in JSON parsing functions, makes it straightforward to query nested FHIR elements and flatten complex FHIR hierarchies into normalized views when needed.

For healthcare organizations looking to share data, the combination of FHIR and data sharing capabilities creates a robust solution for longitudinal patient data sharing. Organizations can load FHIR bundles directly into tables, maintaining multiple FHIR resources in separate tables while preserving the relationships between resources. This approach enables healthcare organizations to create secure data shares containing comprehensive patient records, with temporal data maintained through effective dating and patient history tracked across multiple providers.

Below is an illustration of how that process could look like:

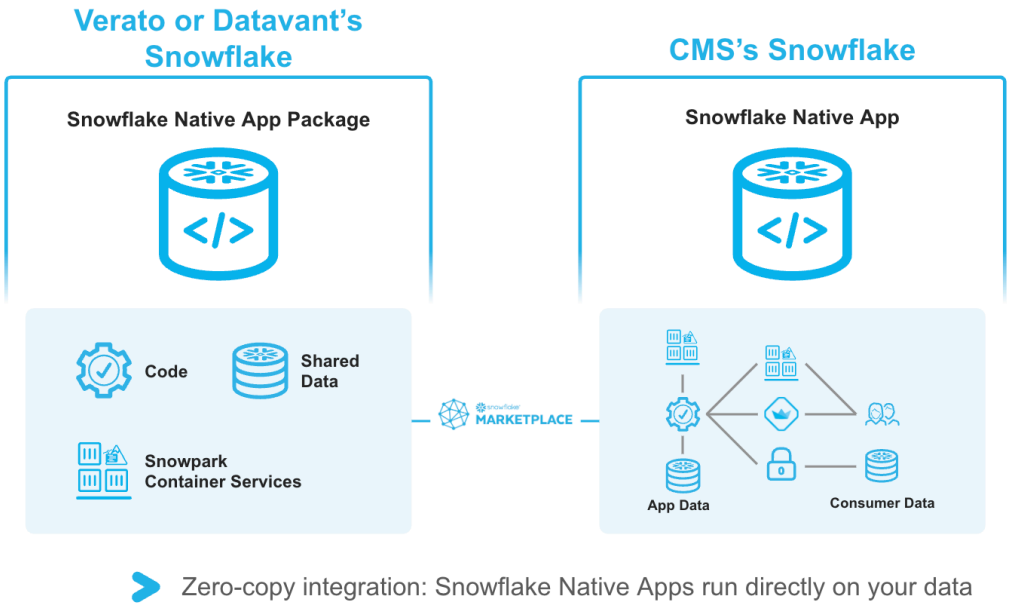



The implementation of this solution offers numerous benefits for healthcare organizations. It significantly reduces data duplication while ensuring secure and compliant data sharing. The cost-effective and

scalable architecture provides real-time access to patient data, making it ideal for various use cases including Health Information Exchanges, Clinical Research Networks, Population Health Management, Care Coordination, and Quality Reporting.

The combination of FHIR and Snowflake data sharing ultimately enables healthcare organizations to share standardized data efficiently while maintaining patient data lineage and enabling real-time analytics. This approach not only reduces infrastructure costs but also ensures data security and compliance while supporting broader interoperability initiatives in healthcare. As healthcare organizations continue to focus on improving data sharing and interoperability, this solution provides a scalable and secure foundation for managing and sharing patient data across the healthcare ecosystem.

## Securing the Future of Interoperability: Snowflake Native Apps for Identity, Privacy and Scale




As CMS seeks to advance digital health interoperability, native applications offer a modern, secure, and scalable approach to deploying critical data services directly within the CMS Snowflake environment.

By running Native Apps, CMS can:

- **Eliminate Data Movement (Zero-ETL Architecture):** Native Apps operate entirely within CMS's existing Snowflake instance, removing the need to extract, transfer, or replicate data—minimizing latency, cost, and risk.

- **Enhance Trust and Data Security:** All data remains within CMS's security boundary, preserving full control and governance over sensitive beneficiary and provider information, while supporting HIPAA and federal compliance frameworks.

- **Enable Cross-Cloud and Multi-Tenant Flexibility:** Snowflake Native Apps run consistently across AWS, Azure, and Google Cloud, allowing CMS to scale capabilities and collaborate with

stakeholders regardless of their cloud infrastructure.

- **Simplify Operations with Composable, Infrastructure-Free Apps:** Native Apps allow CMS to rapidly deploy and iterate on data services—such as patient identity resolution, record linkage, and quality measure computation—without managing infrastructure, containers, or DevOps pipelines.

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## Strategic Snowflake Native Apps Aligned to CMS Priorities

In the following two sections we provide solution briefs from our partners:

- **Verato** provides next-generation **Enterprise Master Patient Index (EMPI)** capabilities, enabling CMS to accurately resolve and unify patient identities across disparate data sources—laying the foundation for longitudinal patient records, coordinated care, and analytics and AI at scale.

- **Datavant** offers **privacy-preserving tokenization** natively within Snowflake, allowing CMS and partners to securely link datasets without exposing personally identifiable information—critical for enabling privacy-compliant research, public health, and data exchange efforts.

## Identity Resolution is Foundational to Healthcare Transformation

*This section was drafted by **Verato Inc.** in partnership with Snowflake. Verato is headquartered at 1751 Pinnacle Drive, Suite 1700, McLean, VA 22102. More information can be found at verato.com.*

*This section is a general response to the entire RFI and specifically applies to Items PA-3, PA-4, PC-5, PC-7, PC-8, PC-14, PR-7, PR-8, PR-9, PR-10, TD-1, TD-3, TD-6, TD-7, VB-2, VB-3a, VB-11, VB-12, VB-14.*

**Introduction**

Identity resolution is more than a data quality initiative—it is the foundation for every healthcare transformation effort that depends on accurate, connected, and complete data. In a health care system where fragmentation across EHRs, CRMs, analytics platforms, and cloud environments are widespread, precisely resolving identity is the essential first step toward achieving true interoperability, improving outcomes, and enhancing operational efficiency.

Critically, identity resolution must go beyond just patients—it must accurately identify and link data for all individuals involved in care delivery and coordination, including caregivers, family members, and healthcare professionals. Whether someone is receiving care, providing it, managing benefits, or supporting a loved one, the ability to reliably match and connect data to the right person is foundational.

This challenge spans the entire healthcare ecosystem: from inpatient and ambulatory settings to long-term care, health insurance, health information exchanges (HIEs), and health technology vendors. Without a unified, trusted identity for each individual, efforts in population health, care coordination, and digital transformation are fundamentally compromised.

### 1. Interoperability Depends on Identity

FHIR, TEFCA, and cloud-native architectures are designed to facilitate data sharing across healthcare systems, but interoperability without robust identity resolution only exacerbates the existing chaos. Identity is the key that enables disparate systems and sources to speak the same language about the same individual or entity, ensuring that data exchanged is linked to the correct person.

HIMSS (Health Information Management Systems Society) defines four levels of interoperability, each of which plays a critical role in enabling seamless data exchange across the healthcare ecosystem. However, none of these levels can function effectively without a solid identity foundation to ensure that the right person is connected to the right data.

The four levels of HIMSS interoperability[1]:

1. Foundational: Establishes the inter-connectivity requirements needed for one system or application to securely communicate data to and receive data from another
2. Structural: Defines the format, syntax and organization of data exchange including at the data field level for interpretation
3. Semantic: Provides for common underlying models and codification of the data including the use of data elements with standardized definitions from publicly available value sets and coding vocabularies, providing shared understanding and meaning to the user
4. Organizational: Includes governance, policy, social, legal and organizational considerations to facilitate the secure, seamless and timely communication and use of data both within and

[1] https://legacy.himss.org/resources/interoperability-healthcare

between organizations, entities and individuals. These components enable shared consent, trust and integrated end-user processes and workflows

While modern systems have achieved much of the above, an essential foundational level is still missing—an Identity Interoperability level. This level, between Semantic (Level 3) and Organizational (Level 4), would focus on accurately identifying which individual the data is associated with and ensuring that the receiving system can match this data to the correct individual. Without identity resolution at this stage, data exchange is meaningless; the information may flow, but it cannot be utilized effectively, defeating the purpose of interoperability. Unlike Structural (level 2) and Semantic (Level 3) where most of the interoperability burden falls on the sender, Identity Interoperability largely falls on the receiver to have accurate person and provider matching in their technical ecosystem in order to match and land the incoming data on the correct records within their system(s).

Although some progress has been made in standardizing patient demographic data across systems, the challenge remains significant. Past efforts have mostly focused on ensuring that demographic data (e.g., name, address, phone number) is collected consistently and in a structured way. However, this has seen limited success due to several key challenges:

- While having a more complete and structured set of demographic data can improve matching algorithms, this only works as long as that data remains stable. Minor changes, such as a new address or phone number, can disrupt matching. More significant changes, like a change in last name due to marriage or divorce, or simultaneous changes to multiple attributes, make accurate identity matching even harder.
- Patients often hesitate to provide certain demographic details—such as Social Security Numbers (SSNs)—due to privacy concerns, especially in light of frequent security breaches. This resistance limits the ability of matching algorithms to reliably link individuals across systems.
- The matching technology most commonly used today relies heavily on accurate data input. However, data entry errors, such as typos or missing information, can significantly hinder matching accuracy. A mistake in a key field (e.g., a typo in date of birth) can break the entire matching process, while issues in other fields (e.g., address or phone number) can still cause errors—especially when multiple fields are inaccurate simultaneously.

To address these challenges and truly unlock the potential of interoperability, an Identity Interoperability layer needs to be established. This layer would ensure that the person behind the data is correctly identified and matched across systems and organizations, even in the face of demographic changes or incomplete data. The success of all healthcare transformation initiatives and modern health information data exchange hinges on this foundational layer. Without it, the promise of seamless, secure, and actionable healthcare data exchange will remain an unfulfilled aspiration, leaving healthcare organizations unable to fully leverage the data exchanged.

Examples:

- Sending a patient's comprehensive, well-structured CDA document, fully coded (with the correct vocabularies and terminologies) to a neighboring provider is meaningless if the receiving EHR can't match the identity because of slight discrepancies in demographic information between the CDA and what is on file at the receiving organization. Without accurate identity resolution, the data is essentially unusable, no matter how well-structured or standardized it is.
- A TEFCA query to retrieve more information about a patient is only useful if it returns meaningful data. In reality, inconsistent identity matching across nodes often leads to no results, simply because the query's demographic details don't perfectly match those stored at other nodes. These small discrepancies can render the query useless, undermining the entire value of data sharing.

- A data lake or warehouse only holds value if it can accurately identify that the patient in the EHR is the same person in the Customer Relationship Management (CRM), HIE, and call center systems. Without robust identity resolution, the data lake risks fragmented and disconnected records, rendering it ineffective.
- A health insurance provider needs to aggregate claims, clinical, and other data from multiple healthcare facilities and other data trading partners to fully understand the members they support the care for. However, discrepancies in patient identity across all the sending systems and organizations result in incorrect matches. This leads to duplicate claims, delays in reimbursement, and erroneous denials or approvals. Without accurate identity resolution, the insurance provider risks significant operational inefficiencies, financial loss, and patient dissatisfaction, as the data being processed may be tied to the wrong individual or treatment episode.

### 2. Accurate Identity = Accurate Insights

Without a trusted identity foundation, patient and provider records become fragmented across systems, leading to data silos, duplication, and critical knowledge gaps. Fragmented data undermines the very initiatives that are central to improving healthcare outcomes: from enterprise analytics to population health management, care coordination, and public health reporting. When identity resolution is lacking, missing links between disparate systems and unresolved duplicate records within individual systems result in data integrity issues, making it impossible to trust the insights derived from that data.

These data gaps and inaccuracies have far-reaching consequences. Enterprise analytics designed to improve operational efficiency or drive strategic decisions are compromised when the data doesn't represent a unified, accurate picture of each individual. For example, population health initiatives—which rely on accurate, connected data to monitor and manage patient cohorts—are fundamentally impaired when identity mismatches prevent systems from recognizing the same individual across care episodes or healthcare providers. This leads to incomplete patient profiles, which hinder effective care coordination and make it difficult to identify at-risk populations.

Additionally public health and quality reporting depend on accurate, holistic data to track disease trends, measure healthcare quality, and guide policy decisions. When patient records are fragmented, this data can be misleading, leading to poor decision-making at both the clinical and policy levels.

Ultimately, without accurate identity resolution, healthcare systems are unable to fully leverage their data to improve patient care, optimize operations, or drive innovation. The ability to connect and unify patient and provider data across systems is foundational to realizing the full potential of digital health transformation.

#### Examples

- It's impossible to accurately track readmission rates, health equity, or cost-of-care trends if the same patient is counted as three different individuals in your data. For example, if a patient has multiple records due to an address change or a system mismatch, it will distort performance metrics, making it difficult to identify opportunities for improvement in care coordination or resource allocation. This fragmentation can lead to under estimating readmissions or misunderstanding health disparities, ultimately hindering efforts to reduce costs and improve patient outcomes.
- Fragmented data can cause false negatives in analytics, leading to missed opportunities in patient care. For instance, a care gap report might suggest that a patient missed an important lab test, triggering unnecessary follow-ups or interventions. However, the lab result may already be documented in a duplicate record from another encounter or location. This results in wasted time

and resources while also potentially delaying the timely care the patient needs. Such errors not only lead to operational inefficiencies but can also impact clinical decision-making.

- In population health management, the goal is to identify trends and at-risk groups. However, if patient records are fragmented or not linked correctly, it's difficult to accurately assess chronic disease prevalence, risk factors, or the effectiveness of health interventions. For example, if a patient with diabetes is counted as two different individuals due to inconsistent records, it may skew the health data for a particular community, leading to misdirected public health interventions or inefficient allocation of resources.
- Accurate public health reporting is vital for tracking disease outbreaks and guiding national health policies. Fragmented data can create misleading reports that fail to accurately represent the true state of public health. For example, during a flu outbreak, inaccurate data might suggest that there are more cases than in reality, leading to misinformed policies or incorrect public health response efforts. Addressing identity fragmentation ensures that public health data is accurate, enabling timely responses and more effective interventions.

### 3. Better Identity Enables Better Outcomes

When clinicians and care teams have a 360-degree view of a patient, powered by trusted identity data, they are positioned to make faster, safer, and more informed decisions. This comprehensive view allows healthcare providers to access a patient's complete medical history, including past treatments, allergies, medications, and any other critical health information, all from one unified source. This not only enhances the accuracy of clinical decisions but also enables real-time collaboration across different providers and specialties.

A complete and 360-degree view of all of a patient's data is more than just an aggregation of individual facts; it allows clinicians to personalize care by understanding the full scope of a patient's needs. Whether it's addressing chronic conditions, managing mental health, or coordinating multiple specialists, this holistic approach makes patient care more tailored and effective, leading to better clinical outcomes. By having all relevant information at their fingertips, clinicians can reduce the risk of medical errors, avoid redundant tests, and prescribe the most appropriate treatments from the outset.

When clinicians and care teams have a 360-degree view of a patient—powered by trusted identity data—they can make faster, safer, and more informed decisions. It also empowers more personalized patient outreach, proactive engagement, and reduce care disparities.

#### Examples

- When care teams have a complete, unified view of a patient—not only from the local EHR but also across the entire healthcare ecosystem, including external partners and data shared through TEFCA queries—they are equipped to deliver whole-person care. This 360-degree view ensures that clinicians make well-informed decisions, considering every aspect of a patient's health. Whether it's medical history from different care settings, clean non-duplicated records in their local EHR, or critical social data from community health partners, a connected ecosystem enables the delivery of personalized care that's more comprehensive and effective. For instance, knowing that a patient with diabetes has recently seen a specialist or undergone a procedure outside the primary care office ensures that all aspects of care are accounted for, reducing duplication and preventing harmful care gaps.
- Integrating nonclinical data, such as social determinants of health (SDoH) data, into a patient's record allows clinicians to better understand the broader factors influencing a patient's well-being—beyond just medical history. For example, if a patient has limited access to transportation, a healthcare system could proactively offer ride-share credits for routine appointments, ensuring the patient can attend preventive care visits. This reduces the likelihood of missed appointments

and costly ER visits due to neglected health concerns. By addressing these non-clinical barriers, healthcare providers can take more holistic action and improve patient engagement, reducing overall healthcare costs in the long run.

- A patient recovering from surgery may have a high risk of hospital readmission if they do not have adequate home support. By matching social data (such as whether the patient lives alone or has access to caregiving support) with their medical record, care teams can ensure the patient receives extra home health visits, or additional services like meal delivery or post-surgery physical therapy. This personalized care helps prevent complications and reduces readmissions, while also improving the patient's overall recovery process.

### 4. Operational Efficiency Requires Trusted Identity Data

From automating workflows to powering AI-driven decision-making and enhancing call center routing, trusted identity data is the bedrock that enables healthcare systems to operate more efficiently by removing friction and eliminating waste. When patient and provider data is accurate, unified, and up-to-date, the entire healthcare ecosystem becomes more agile, reducing unnecessary steps and manual intervention.

For example, a clean, unified record eliminates the need for duplicate data entry and reduces the time spent by administrative staff validating patient information. When every healthcare system and application is connected to a trusted identity, routine tasks can be automated without manual lookups, minimizing the risk of errors and enhancing the speed and efficiency of patient care.

Examples

- In a hospital or clinic setting, manual entry of patient information is time-consuming and prone to errors. If a system doesn't recognize that the patient in the ER is the same patient that has been treated by the cardiology department, staff will have to manually search multiple systems, re-enter data, or even call the patient to confirm details. With trusted identity resolution, this process is automated. The system seamlessly connects patient records across the enterprise, reducing administrative burden and allowing staff to focus on patient care.
- In healthcare call centers, identity resolution helps to ensure that patients are routed to the right department and provider based on their needs, significantly reducing the number of transfers and wait times. For instance, if a patient calls to inquire about their recent blood test results, the system instantly recognizes their identity, pulling up the correct medical record, so the call can be routed directly to the correct provider or department, bypassing administrative staff and reducing patient frustration.

### 5. Regulatory & Compliance Risks Shrink with Trusted Identity Data

Duplicate or mismatched patient records not only create operational inefficiencies but also expose healthcare organizations to significant regulatory and compliance risks. Without trusted identity data, healthcare organizations face the very real danger of violating privacy laws, incurring financial penalties, and damaging their reputation.

Examples

- Mismatched patient records pose a direct threat to HIPAA compliance. If a healthcare organization cannot accurately identify and link patient records across different systems, sensitive information could be disclosed to unauthorized individuals or entities, violating patient privacy

protections. Trusted identity resolution ensures that patient data is correctly associated with the right person and remains confidential, reducing the risk of accidental exposure or unauthorized access.

- Healthcare providers are required to maintain accurate, transparent audit trails to demonstrate compliance with HIPAA and other healthcare regulations. Duplicate or fragmented patient records make it difficult to maintain a consistent and accurate record of data access and usage, which can result in audit failures and incorrect or incomplete patient rights such as 'accounting of disclosures.
- Consent management is a cornerstone of HIPAA compliance and health care operations, particularly in relation to data sharing across healthcare providers, insurers, and other third parties. If patient identities are not properly matched, the system may not be able to accurately track whether a patient has consented to share their medical records with other entities. This data mismatch can result in violations of patient consent.
- Duplicate or mismatched records often lead to incorrect billing and insurance claims. This may lead to penalties delay claims processing, affecting revenue cycles.

### 6. Training & Use of Responsible AI Requires Complete & Trusted Datasets

In healthcare, Artificial Intelligence (AI) is revolutionizing everything from diagnostics and clinical decision support to predictive analytics and personalized care. However, AI's ability to deliver meaningful, accurate, and equitable results depends not only on how it's trained and the quality of the training data, but also on the quality and completeness of the data it uses once trained. Trusted identity resolution is the foundation for both training responsible AI and ensuring that AI systems in healthcare have access to the most accurate and comprehensive data possible. Without resolving patient identities consistently across systems, healthcare AI risks producing results that are biased, incomplete, or inaccurate, which can lead to poor outcomes.

To harness AI's full potential in healthcare, organizations must ensure that patient and provider data is correctly linked, up-to-date, and accurate—across the entire ecosystem of clinical systems, care coordination tools, and third-party data. Only then can AI systems make safe, fair, and data-driven decisions that improve patient care, optimize workflows, and reduce healthcare costs while maintaining regulatory compliance. Whether in training AI models or utilizing them in real-world applications, trusted identity resolution is essential for the responsible and effective use of AI in healthcare.

Examples:

- An AI system in a hospital uses patient records to help clinicians make diagnostic decisions. However, due to duplicate records or inaccurate identity resolution, the AI draws on fragmented or conflicting data about the same patient (e.g., an incomplete medical history, allergy records, or lab results). The AI might suggest an incorrect diagnosis or fail to identify critical drug interactions, allergic reactions, or undiagnosed conditions. This can lead to misdiagnosis, ineffective treatment plans, and potentially dangerous health outcomes.
- A telemedicine platform uses AI for initial diagnosis and care recommendations based on data pulled from various sources (e.g., EHRs, remote patient monitoring devices). However, if there are mismatched or duplicate patient records between the telemedicine platform and systems it pulls from, the AI might process incorrect information or base a recommendation on incomplete information.

### 7. Identity Verification/Proofing is an Important & Relate Step

In healthcare, identity verification and proofing are essential steps in ensuring the accuracy, security, and integrity of patient and provider data across the healthcare ecosystem. Effective identity verification allows healthcare organizations to confirm that the individual accessing services or data is who they say they are—whether the individual is a patient, caregiver, family member, or healthcare professional.

The recent federal regulations in the U.S., such as the 21st Century Cures Act and TEFCA (Trusted Exchange Framework and Common Agreement), emphasize the need for secure and accurate identity verification as part of the movement toward interoperable health data systems. As healthcare organizations adopt advanced digital solutions, effective identity proofing ensures that the right data reaches the right individuals, whether they are clinicians, patients, or third-party partners. Without robust identity verification, the integrity of healthcare data systems is compromised, which can lead to fraud, misidentification, and errors in care delivery.

However, the process does not end with identity verification—the verified identity must seamlessly flow throughout the entire healthcare ecosystem. Too often, identity verification is treated as a standalone process, and valuable data, such as patient demographics obtained during verification, is discarded after the verification step. This approach overlooks the critical role that this verified data plays in ensuring consistent, accurate identity resolution across multiple systems.

Take, for example, a typical use case for patient identity verification: the creation of a portal account. Identity verification is required at this stage to ensure that the person accessing the portal is who they claim to be. However, this verification should not be the final step. The verified demographics, such as the patient's name, date of birth, and contact information, should not only be retained but also used to match the portal account to the patient's correct medical record in the system. This ensures that the individual interacting with the system is correctly tied to their health data.

The verified identity also needs to flow across the entire healthcare ecosystem. When identity verification and resolution work together in a holistic way, all systems involved in patient care can leverage the same trusted identity. Only by ensuring that verified identity data is continuously shared, updated, and referenced across the entire healthcare ecosystem can organizations fully realize the benefits of accurate, reliable, and interoperable healthcare data. This alignment solves larger and more complex use cases than just the initial identity verification needs.

Examples:

- If a patient successfully completes identity verification but is mistakenly linked to a different patient with similar demographics, the entire process fails. Although the individual's identity is verified, the system erroneously connects them to the wrong medical record, which not only causes significant patient frustration but also risks the exposure of protected health information (PHI).
- When a healthcare professional completes identity verification but is incorrectly linked to a different user account, there is a significant risk that inappropriate access rights will be granted. This not only compromises the security of patient data but could also lead to unauthorized access to sensitive health records, which may result in legal and compliance violations. Without proper identity resolution, the professional could be given access to areas they should not, increasing the risk of fraud, data misuse, and potentially violating various regulations.

### 8. Longitudinal Views, Not Encounters, Lead to Better Health Outcomes

In the traditional healthcare model, patient data is often fragmented, siloed, and tied to specific episodes of care or encounters—meaning that information is only collected or updated during a patient's visit to a healthcare provider. However, this episodic approach fails to capture the full continuum of care for a patient and limits the ability to make holistic, data-driven decisions that can improve both health outcomes and operational efficiency.

Building a longitudinal view of patient health, where data is continuously gathered and integrated across all encounters, allows healthcare organizations to gain a complete picture of a patient's history, behaviors, and ongoing needs. This longitudinal approach ensures that all touchpoints in a patient's healthcare journey—from primary care visits to hospital admissions to wellness checkups—are captured and linked together to form an accurate, real-time, and comprehensive health record.

However, achieving this comprehensive view requires more than just aggregating data from different encounters; it demands consistent identity resolution to ensure that a patient's health information is consistently and correctly linked across systems, organizations, and over time. Given the longitudinal nature, the matching solution must be able to deal with the patient's demographics changing over time and with life events (such as marriage and divorce), as well as misaligned attributes from systems that may collect different demographics and identifiers. Without this foundational step, fragmented, inconsistent, or inaccurate data can lead to missed opportunities for early intervention, care coordination breakdowns, and inefficient operations.

Examples:

- A patient has unresolved duplicate records within the same EHR due to slightly different demographic data (e.g., "John A. Smith" vs. "Jonathan Smith"). Without proper identity resolution, each record appears as a sperate patient and not included in the comprehensive longitudinal view.
- A patient is treated at a hospital outside their primary health system. The data is share back with the primary system, but the data from the external EHR isn't matched due to variations in demographic fields.
- An HIE struggles to build out a longitudinal view because its dozens of contributing health system each send in slightly different demographics and there is no common identifier between systems.

### 9. Patient Experience Suffers without Trusted & Complete Identity

Identity resolution is a cornerstone of improving patient experience in healthcare transformation. By establishing a single, accurate, and unified patient identity, healthcare providers can deliver truly personalized care, streamline administrative processes, and ensure seamless continuity across multiple care settings. This foundational capability not only supports better health outcomes but also fosters increased patient satisfaction and stronger, more trusting patient-provider relationships.

Today's patients expect effortless, secure access to their complete health information—whether through patient portals, mobile apps, or connected devices. Effective identity resolution ensures that all relevant data, from lab results to medication histories, is accurately linked to one comprehensive patient profile. This prevents confusion caused by fragmented or duplicate records, and delivers a consistent, holistic view of their health journey. Trusted identity also enables healthcare organizations to tailor communications, reminders, and care workflows to the individual's unique needs, enhancing engagement and adherence.

Without robust and reliable identity resolution, patients risk encountering incomplete, inconsistent, or incorrect information due to duplicate or unlinked records across and within systems. Such gaps can lead to patient confusion, frustration, erosion of trust in the healthcare provider, and even raise privacy and security concerns when data is mismatched or exposed improperly.

Examples:

- A patient logs into their healthcare provider's portal to review details of a recent emergency department visit. However, because the emergency department was unable to locate the patient's existing record, a new duplicate record was created for that visit. As a result, the patient's portal does not display the emergency visit information, leading to confusion. When the patient calls to ask about the missing data, they struggle to understand why their records are incomplete or inconsistent, undermining trust and satisfaction.
- A patient is referred to a specialist within the same health system as their primary care provider (PCP). Expecting seamless coordination, the patient logs into the organization's portal but cannot find any information from the specialist visit. Unbeknownst to the patient, the specialist uses a different EHR system, and the encounter data was never linked to the patient's primary medical record. This fragmentation disrupts care continuity and leaves the patient feeling disconnected from their own healthcare journey.

**Conclusion**

Without a trusted identity foundation, data is limiting, even dangerous. With trusted identity foundation, data becomes powerful, insightful, meaningful, and actionable. Identity resolution is not just a technical requirement—it is the essential foundation for transforming healthcare delivery, improving patient outcomes, and driving operational excellence. Accurate and trusted identity enables seamless data integration across fragmented systems, empowering clinicians, patients, and organizations with a complete and unified view of health information. Without it, the promise of interoperability, personalized care, efficient workflows, and advanced analytics remains out of reach. As healthcare continues to evolve—with increasing reliance on digital tools, data sharing networks, and AI—addressing the lack of robust identity resolution capabilities is critical to building a healthcare ecosystem that is connected, trusted, and patient-centered. Ultimately, resolving identity challenges unlocks the full potential of healthcare data and enables transformation, delivering better care experiences and healthier populations.

It is vital to ensure that all parts of the health care ecosystem have accurate patient matching as part of their essential infrastructure. This will only continue to increase in importance as health systems, payers, and health tech companies move toward smarter, more connected care.

Suggestions

- Encourage the adoption and use of better and proven record matching technology. There are newer matching solutions that are proven to overcome all the challenges mentioned above. Commonly referred to as "referential matching," these solutions incorporated a trusted and vetted set of reference data for the nation-wide population, that is used to overcome changes to a person demographics overtime, and the missed-aligned attribute problem; where one record might have name, DOB and address while the other record might have name, SSN and phone number. These solutions also typically include more advance probabilistic approaches than the typical EHR and other certified health information technology on the market today.

Incorporate requirements into certified health information technology to have accurate and advanced matching capabilities. The majority of the existing four levels of the HIMSS defined interoperability levels

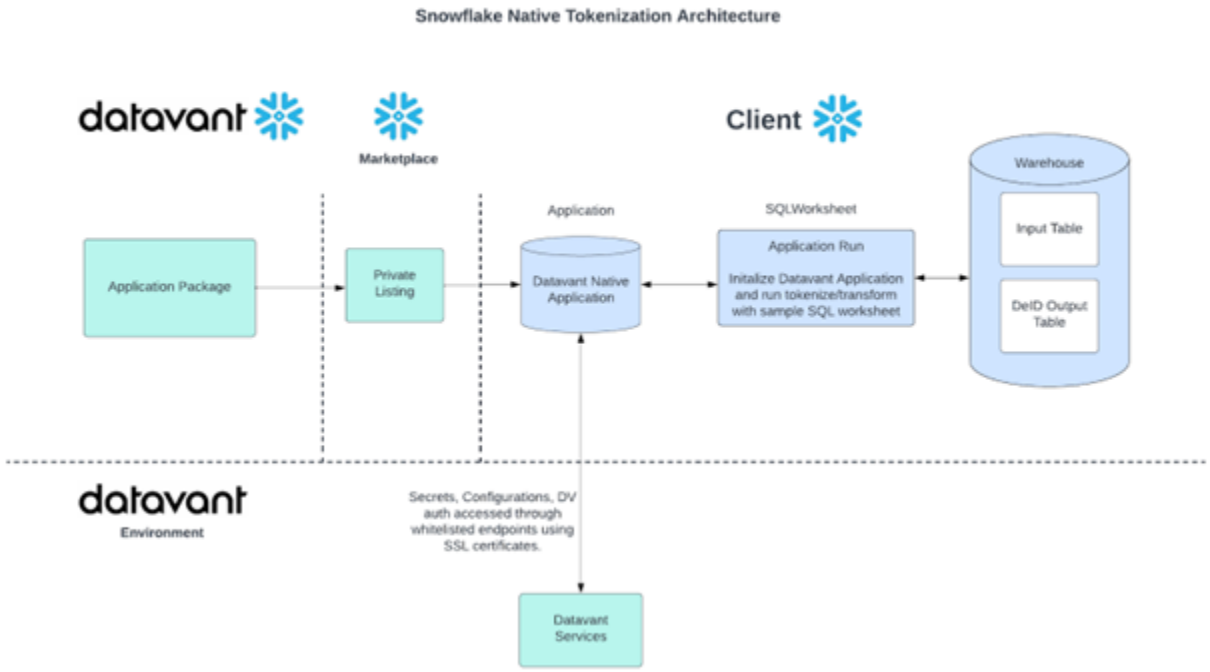
have already been addressed with requirements to certified health technology. However, little has been done to require certified health information technology to address the missing 'Identity Interoperability' level. The robustness and accuracy between different vendors vary widely, leading to very inconsistent results and differing levels of fragmentation depending on what technology vendor does the matching. That needs to change to require certified health information technology vendors to take up the lack of identity interoperability in the current systems and data exchanges. Accuracy of the patient record matching is just as vital as Structural and Semantic interoperability; in fact, it is more important. What is the point of sending data if the most fundamental item, what patient the data is for, is not understood by the receiving system.

## Tokenize, Link, and Analyze Healthcare Data Natively in the Cloud

*This section was drafted by Datavant Inc. in partnership with Snowflake. Datavant is headquartered at 44 Montgomery Street, San Francisco, CA 94101. More information can be found at datavant.com.*

### Overview

Datavant and Snowflake have partnered to deliver a fully native, privacy-preserving data linkage solution that empowers healthcare and life sciences organizations to tokenize, transform, and analyze data—all from within the Snowflake environment. This joint offering eliminates data movement, simplifies compliance, and enables real-time collaboration using Snowflake's Data Cloud and Clean Room technologies.




### Key Benefits

- Native Tokenization: Tokenize and transform PHI in-place via the Datavant Native App—no data exports.
- Faster Time to Value: Reduce ETL complexity and accelerate insights using in-platform processing.
- Privacy-Preserving: De-identify and re-encrypt sensitive data for secure multi-party use cases.
- Governed Collaboration: Use Snowflake Clean Rooms to share and analyze data without exposing raw PII/PHI.
- Scalable and Secure: Leverage Snowflake's performance, compliance, and governance at scale.

### Solution Capabilities

- Datavant Capabilities: Tokenize PHI/PII, Transform tokens between partners, Support for real-world health data, Native App via Snowflake Marketplace

- Snowflake Capabilities: Secure compute, storage, and governance, Clean Rooms for secure collaboration, Snowflake Cortex for AI/ML insights, Native data sharing and multi-party access

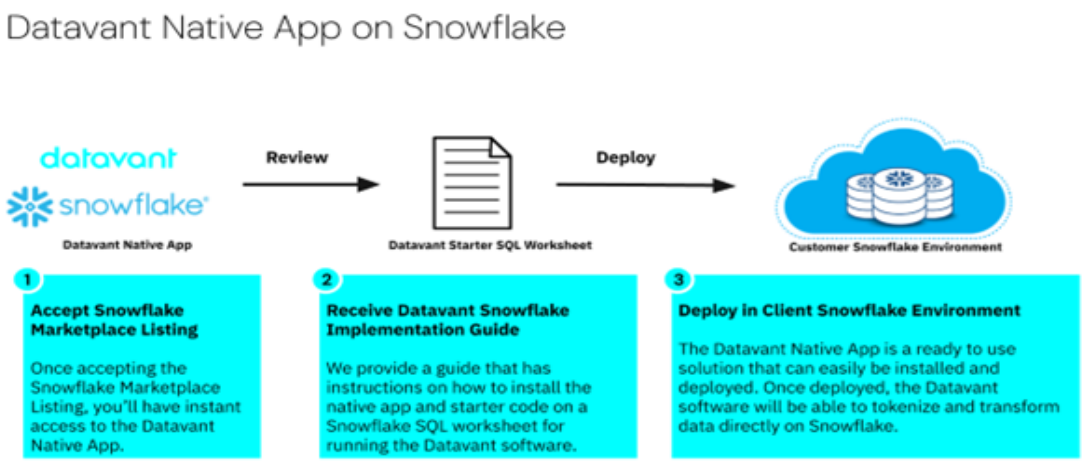
### Primary Use Cases

- Patient Journey Mapping across providers and data sources
- Clinical Trial Feasibility using claims and EHR linkage
- Next Best Action Models for commercial targeting
- Rare Disease Detection via tokenized third-party data
- FDA Submissions and RWE Studies
- Adverse Event Monitoring and Pharmacovigilance

### Example Outcome

A national health system linked over 50 million tokenized records in under 6 weeks using Datavant's Snowflake Native App, reducing manual data wrangling and enabling cross-partner analytics in a HIPAA-compliant manner.

### Implementation Overview




1. Get Access via Private Listing: Share your Snowflake Account ID with Datavant to receive access.
2. Deploy App in Snowflake: Install the Datavant Native App from the Snowflake Marketplace.
3. Create Tokenization Configs: Use Datavant's configuration guides and portal for setup.
4. Grant Permissions: Assign Snowflake privileges and external access integrations.
5. Run SQL-Based Tokenization: Use Snowflake worksheets with provided sample scripts.
6. Transform Tokens Across Partners: Re-encrypt data for partner sharing using transform() functions.
7. Analyze or Share Data: Leverage Clean Rooms, Cortex, or Snowflake-native SQL tools.

### Why This Matters

Together, Snowflake and Datavant simplify the hardest part of healthcare data interoperability: privacy-preserving data linkage. With native functionality, reduced risk, and faster insights, organizations can move from siloed datasets to collaborative analytics—without sacrificing compliance or control.

# Appendix- Data Governance and Security with Snowflake

## Snowflake Horizon

Snowflake Horizon is a comprehensive, built-in data governance and discovery solution within the Snowflake AI Data Cloud. It provides a unified control plane for organizations to manage, protect, and gain insights from their structured, semi-structured, and unstructured data, applications, and models across various cloud environments. Essentially, Horizon aims to simplify the complexities of data governance at scale, enabling secure collaboration and accelerating the value derived from data.

**Key Features and Capabilities:**

**Unified Governance and Compliance:** Horizon offers a centralized approach to enforce data governance policies, ensuring compliance with regulatory standards. This includes capabilities for:

- *Automated Sensitive Data Discovery and Tagging*: Leveraging AI, Horizon can automatically classify and tag sensitive data (e.g., PII, PCI), streamlining the process of identifying and protecting critical information.
- *Granular Access Controls:* It allows for the centralized application of fine-grained access policies (like row access policies, tag-based masking, and dynamic masking) across databases, schemas, and tables, including open table formats like Iceberg.
- *Data Lineage:* Users can visualize the end-to-end flow of data, understanding upstream and downstream dependencies for impact analysis, root cause identification, and auditing purposes.
- *Data Quality Monitoring:* Horizon provides tools to continuously monitor data quality, allowing for the identification and remediation of issues through built-in and user-defined data metric functions.
- *Audit and Security Monitoring:* It offers continuous risk monitoring, vulnerability management, and robust auditing capabilities to track data usage and ensure secure access.

**Enhanced Data Discovery and Collaboration:** Horizon significantly improves the discoverability and usability of data assets:

- *Universal Search*: A powerful search interface allows users to quickly find data, applications, and models within and outside their organization, often powered by natural language understanding.
- *AI-Powered Descriptions:* Automatically generated descriptions for tables and columns enhance data understanding and discoverability.
- *Internal Marketplace*: Facilitates secure and governed sharing of data products and applications within an organization, promoting internal collaboration.
- *Secure Data Clean Rooms*: Enables privacy-preserving collaboration on sensitive data with external partners or cross-functional teams without moving or exposing the raw data.
- *Synthetic Data Generation*: Allows for the creation of realistic, anonymized copies of sensitive data, expanding opportunities for analysis while maintaining privacy.

**Interoperability:** Snowflake Horizon extends its governance and discovery capabilities beyond Snowflake-native data to include external data sources and open table formats like Apache Iceberg, ensuring a consistent policy model across diverse data ecosystems.

**Benefits for Organizations:**

- ➢ Improved Data Trust and Reliability: By providing robust governance, lineage, and data quality features, Horizon helps organizations ensure the accuracy, consistency, and trustworthiness of their data.

- Accelerated Data Value: Easier data discovery and secure collaboration empower data teams to find, understand, and leverage data more quickly, leading to faster insights and innovation, especially in AI and application development.
- Enhanced Security and Compliance: Centralized controls, automated sensitive data identification, and continuous monitoring reduce security risks and simplify compliance with data regulations.
- Streamlined Data Management: Horizon simplifies complex data management tasks, reducing manual effort and operational overhead.
- Scalability and Efficiency: Built on Snowflake's scalable infrastructure, Horizon handles vast volumes of data efficiently without compromising performance.

In essence, Snowflake Horizon acts as the foundational layer for a well-governed and discoverable data environment, enabling organizations like CMS to unlock the full potential of their data for analytics, AI, and application development while maintaining strong control and compliance.

## Snowflake Security

Snowflake maintains a robust security posture, demonstrated by a comprehensive set of certifications and adherence to leading industry standards, providing CMS with the assurance necessary for handling its sensitive enterprise data. Our platform holds key certifications including ISO/IEC 27001:2022, a globally recognized standard for information security management systems, along with ISO 27017:2015 and ISO 27018:2019, which focus on cloud security and protection of personally identifiable information in public clouds, respectively. These ISO certifications attest to Snowflake's commitment to establishing, implementing, maintaining, and continually improving its information security controls.

In addition, Snowflake undergoes rigorous third-party audits to achieve SOC 1 Type II and SOC 2 Type II attestations. The SOC 1 Type II report provides assurance regarding the internal controls over financial reporting, while the SOC 2 Type II report addresses the security, availability, processing integrity, confidentiality, and privacy of the system. These reports offer valuable insights into Snowflake's operational controls, validating their effectiveness over a defined period.

In addition, as noted above, Snowflake is FedRAMP certified at Moderate and High. Snowflake has achieved FedRAMP High Authorization on AWS GovCloud (US-West and US-East Regions), as well as FedRAMP Moderate Authorization for Snowflake on AWS US East and Microsoft Azure Government. We also have DoD Impact Level 5 (IL5) Provisional Authorization on AWS GovCloud. While the FedRAMP Package ID is typically obtained by authorized federal agencies through USDA Connect or by request via a Snowflake Account Executive, the status itself indicates a robust security posture and compliance with stringent government standards. Snowflake continues to expand its government-focused deployments and pursue additional certifications.

Beyond these major certifications, Snowflake also holds certifications such as HITRUST CSF (through participation in the Shared Responsibility and Inheritance Program), PCI DSS Service Provider Level 1 (for Business-Critical Edition and Virtual Private Snowflake Edition), and Cyber Essentials Plus (CE+) in the U.K. These diverse certifications collectively illustrate Snowflake's dedication to maintaining a highly secure and compliant cloud data platform, addressing a wide array of regulatory and industry-specific security requirements. External security rating bodies like UpGuard have assessed Snowflake's external attack surface, with UpGuard giving Snowflake an "A" security rating, indicating a strong security posture based on continuous monitoring of various risk categories such as website, email, phishing & malware, brand & reputation, and network security.

## Data Privacy

Snowflake's architecture is designed with a strong emphasis on data privacy and compliance with global regulations such as GDPR, CCPA, and HIPAA (where applicable), while also addressing critical enterprise data sovereignty requirements. For GDPR (General Data Protection Regulation) and CCPA (California Consumer Privacy Act), Snowflake provides a robust set of features that enable customers to meet their obligations. This includes granular, role-based access controls (RBAC) and attribute-based access control (ABAC) to ensure that only authorized individuals can access personal data. Dynamic Data Masking allows organizations to protect sensitive information by obscuring it at query runtime without altering the underlying data, offering options for masking, partial masking, obfuscation, or tokenization. Row Access Policies further enhance control by limiting data visibility based on specific conditions, such as user roles or consent status, ensuring data is only accessed for its intended purpose. Audit logging provides a comprehensive record of all data access and modifications, crucial for demonstrating compliance and investigating potential breaches.

Regarding HIPAA (Health Insurance Portability and Accountability Act), Snowflake offers a secure and scalable platform suitable for handling Protected Health Information (PHI) by supporting technical safeguards like end-to-end encryption of data at rest and in transit (AES 256-bit encryption with a hierarchical key model). Snowflake is able to sign Business Associate Agreements (BAAs) with covered entities, which is a fundamental requirement for HIPAA compliance. Customers leveraging Snowflake in healthcare environments can utilize its features, alongside their own administrative and physical safeguards, to maintain compliance with HIPAA's Privacy, Security, and Breach Notification Rules.

For enterprise data sovereignty, Snowflake allows customers to choose the geographical region for their data storage across major cloud providers (AWS, Azure, GCP). This capability is fundamental for organizations needing to keep their data within specific national or regional borders to comply with local laws and regulations. Snowflake's replication capabilities through Snowgrid further enable organizations to replicate data across different regions or clouds while maintaining full control over data placement, ensuring data residency requirements are met even in multi-cloud or hybrid environments.

Snowflake provides various mechanisms to support consent management, data retention, and Data Subject Access Request (DSAR) workflows. While Snowflake itself doesn't directly manage individual user consent forms, its robust data governance features allow organizations to implement their consent management strategies effectively. For example, Row Access Policies can be configured to restrict access to personal data based on recorded consent, ensuring data is only processed for the purposes for which consent was given. Snowflake's native "Time Travel" feature is crucial for data retention, allowing historical data to be retained for a configurable period (up to 90 days in Enterprise Edition and higher). This facilitates regulatory compliance by enabling organizations to retrieve data as it existed at specific points in time, supporting audit trails and legal hold requirements. For DSAR workflows, the combination of comprehensive audit logs, data discovery capabilities (e.g., using Snowflake's Information Schema and object tagging), and secure data sharing allows organizations to efficiently identify, retrieve, and provide personal data to data subjects, or to implement data deletion requests through controlled processes. Partnerships with data governance platforms, such as the OneTrust Consent Management Snowflake Native App available on Snowflake Marketplace, further enhance these capabilities by integrating consent preferences directly with Snowflake's native security controls, streamlining the enforcement of data policies based on real-time consumer choices.